



## NATIONALITY

- Cypriot

## EDUCATION

- PhD. in Computer Science, University of Camerino, 2022, Camerino, Italy
- MSc. in Sustainable Environment and Energy Systems, Middle East Technical University, Northern Cyprus Campus, 2018, Güzelyurt, Cyprus
- MSc. in Computer Science, University of Maryland, 1998, College Park, MD, USA
- BSc. in Computer Engineering, Boğaziçi University, 1994, Istanbul, Turkey

## WORK EXPERIENCE

- **Current, PostDoc:** **Automation and Control** group of the **Dept. of Electronic Systems**, Aalborg University, Aalborg, Denmark.
- **2023 - 2025, Postdoc:** **AI4OR** group of the **Dept. of Materials and Production**, Aalborg University. Research in AI and Autonomous Systems.

### Courses taught:

- Spring 2025 - GBE2 - Global Business Engineering 2 - Forecasting and Inventory Management (half of the semester)
- Spring 2024 - GBE2 - Global Business Engineering 2 - Forecasting and Inventory Management (quarter of the semester)
- Fall 2023 - GBE3 - Global Business Engineering 3 - Basic Optimization and Python Programming (half of the semester)

### Supervising experience:

- Spring 2025 - Global Business Engineering 4, Group Term Project (3 Groups of 6 students)
- Fall 2024 - Global Business Engineering 3, Group Term Project (3 Groups 5-6 students)
- Spring 2024 - Global Business Engineering 4, Group Term Project (1 Group of 6 students)
- Fall 2023 - Global Business Engineering 3, Group Term Project (2 Groups 5+6 students)
- Fall 2023 - (Co-supervising) Mathematics - Economy Term Project (Individual project)

- **2022 - 2023, Instructor:** Dept. of Computer Engineering, Middle East Technical University, North Cyprus Campus, Güzelyurt, Cyprus.

### Courses taught:

- STAS 221 Statistics For Engineers I - Spring 2023
- CNG 223 Discrete Computational Structures - Fall 2022
- CNG 331 Computer Organization / EEE 445 Computer Architecture I - Fall 2022
- CNG 336 Introduction to Embedded Systems Development / EEE 347 Introduction to Microprocessors - Spring 2023
- CNG 495 Cloud Computing - Fall 2022

- **2019 - 2022, PhD. Student:** Dept. of Computer Science, University of Camerino, Camerino, Italy. Researched in Optimization, Drones, WSN, Distributed Systems, Computer Vision, Human Pose Detection, Robotic Arms.

- **2016 - 2019, Teaching Assistant:** Dept. of Computer Engineering, Middle East Technical University, North Cyprus Campus, Güzelyurt, Cyprus. Worked as a teaching assistant and studied for MSc. in Dept. of Sustainable Environment and Energy Systems.

### Courses assisted:

- CNG 111 Introduction to Comp. Engineering Concepts (Python Programming)
- CNG 140 C Programming
- ECO 205 Statistics for Economists
- CNG 213 Data Structures
- CNG 223 Discrete Computational Structures
- CNG 230 Introduction to C Programming
- CNG 242 Programming Language Concepts
- CNG 280 Formal Languages and Abstract Machines
- CNG 301 Algorithms And Data Structures
- CNG 315 Algorithms

- **2007 - 2012, 2015 - 2016 Instructor:** Dept. of Computer Engineering, Near East University, Nicosia, Cyprus. Taught various Computer Engineering courses and researched in Texture Recognition.

**Courses taught:**

- COM 111 Introduction to Computer Engineering
  - COM 121 Discrete Structures
  - COM 132 Introduction to Computers and Programming
  - COM 131/141/142 Introduction to Programming
  - COM 411 Software Engineering
  - COM 416 Computer Networks
- **1996 - 1998, Research Assistant:** Dept. of Computer Science, University of Maryland, College Park. Studied on Duplicate Detection in Large Document Databases in CfAR. Researched in Multimedia Databases Lab, MD, USA.
  - **1993 Summer Training:** Software Engineer, Promax, C programming and UNIX Platform, Istanbul, Turkey.

## PARTICIPATION IN FUNDED PROJECTS

- **2026 - present, D4D:** Drones 4 Defence. Developing AI-based solutions that make it easier to coordinate and manage multiple drones. Currently working as a Postdoctoral researcher
- **2023 - 2025, OASAR:** Optimized and Automated Solution for Maritime Search and Rescue (SAR) Operations. Worked as a Postdoctoral researcher.
- **2019 - 2022, URRA:** Usability of robots and reconfigurability of processes: enabling technologies and use cases, on the topics of User-Centered Manufacturing and Industry 4.0, which is part of the project EU ERDF, POR MARCHE Region FESR 2014/2020–AXIS, "HD3Flab-Human Digital Flexible Factory of the Future Laboratory", coordinated by the Polytechnic University of Marche. Robot Usability and Process Reconfigurability. Worked as a doctoral researcher.

## RELEVANT SKILLS

- **PhD topic (Computer Science):** "Geometry-Based Optimization Heuristics for Region Coverage and Pathfinding in Drone-Based Operations" (**Poster - Thesis**). I proposed a flexible Evolutionary Algorithm (EA) based multi-objective multi-party optimisation scheme for **disaster region communication coverage** and a framework for heuristic drone pathfinding over an optimised charging station grid for "visiting" entities in a "covered region". The **boat rescue** case study is presented as an application of the proposed framework. The thesis involves topics as Graph Theory, Optimization, Combinatorics,  $\mathcal{NP}$ -Completeness, Evolutionary Algorithms, Probability Analysis, Statistical Analysis and Testing, Heuristics, TSP. Coding is done in R language and Linux shell scripting.
- **Reviewer Activities:**
  - Certificate of Excellence from Certified Peer Reviewer Course, Elsevier Research Academy, awarded May-18-2021
  - I have reviewed for:
    - \* Aging and Disease, International Society on Aging and Disease
    - \* Computer Communication, Elsevier
    - \* Expert Systems, Wiley
    - \* International Journal of Pattern Recognition and Artificial Intelligence, World Scientific
    - \* Physica A: Statistical Mechanics And Its Applications, Elsevier
    - \* Sustainable Cities and Society, Elsevier
    - \* Wireless Communications and Mobile Computing, Hindawi-Wiley
    - \* Internet of Things, Elsevier
    - \* European Journal of Industrial Engineering, Inderscience
    - \* International Journal of Vehicle Systems Modelling and Testing, Inderscience
    - \* 15<sup>th</sup> IFAC Workshop on Intelligent Manufacturing Systems - IMS 2025
    - \* The Journal of Supercomputing, Springer
    - \* Sensors, MDPI
    - \* Statistics in Medicine, Wiley
    - \* Transportation Research. Part E: Logistics and Transportation, Elsevier
- **Pedagogical formation for university education:** I attended and got **certificate for Problem Based Learning course from the Aalborg University**, Denmark in 2023.
- **Analytical reasoning:** I am good at seeing the parallelism between diverse topics and finding research gaps. I can classify/categorize existing approaches into an organized literature review.
- **Strong problem-solving attitude:** My multidisciplinary background and creative nature gives me flexi-

bility in proposing novel solutions.

- **Strong communication and English language skills (written and spoken):** I have about 10 years (7 as instructor + 3 as TA) of teaching experience at the institutions where English was the language of education. I stayed about 5 years in the USA.
- **Teamwork:** I have many experiences in various collaborative research projects and faculty board meetings focused on undergraduate engineering curriculum design.
- **The ability to properly report, organize and publish research data:** I have publication experiences in various Computer Science fields like Biometrics, Networks, and Heuristics.

## ADDITIONAL SKILLS

- Passionate hobbyist for Robotics, TinyML, and IoT projects with Micro:bit, Arduino, RaspberryPi, DJI-Tello, Crazyflie 2.X, and JetsonNano.
- Developed prototype webcam based human pose detection software (Python, OpenCV) as part of the collaborative project (Urrà: Robot Usability and Process Reconfigurability) with the Università Politecnica delle Marche.
- Basic electronics, soldering, building breadboard prototypes. Practical experience with various sensors (ultrasonic, PIR, etc...) types. Practical experience with step motors, servos (Developed Arduino sketch for running multiple servos at the same time for "smooth" robot arm movements: <https://github.com/kk-1/robot-arm-demos>).
- Linux, R, OpenCV, Python, LaTeX, Docker
- LinkedIn Skill Assessment badges for:
  - "R Programming Language", 2022
  - "Python Programming Language", 2022
  - "Machine Learning", 2022
- Datacamp Statement of Accomplishment for:
  - "Data Science R Basics", 2022
  - "Spatial Statistics in R", 2022
  - "Network Analysis with R Track", 2022
- Advance level (not certified) of Italian
- Collaboration with Electrical Engineering Department for developing a fractal antenna as a part of my MSc. thesis titled "Energy Efficient Routing With Directional Antennas In Wireless Sensor Networks"(<https://github.com/kk-1/metu-msc-thesis>)

## PROFESSIONAL MEMBERSHIPS

- ACM
- IEEE: Member of the Evaluation Committee in [IEEE Duino 2024-2025](#) (Awarded [Certificate of Appreciation](#)).
- IEEE Computer Society

## PRESS/MEDIA

- 13/05/2022 - Faculty of Engineering and Natural Sciences Webinars, Kadir Has University, Turkey:  
[Geometry-Based Optimization Heuristics for Region Coverage and Pathfinding in Drone-Based Operations](#)

## PUBLICATIONS

- [1] Kemal Ihsan Kilic, Aurélien Desoeuvres, Casper Bak Pedersen, Alex Elkjær Vasegaard, and Peter Nielsen. Adaptive artificial potential field method for small autonomous vehicles. *Robotics and Autonomous Systems*, 198:105364, 2026. <https://doi.org/10.1016/j.robot.2026.105364>.
- [2] Kemal Ihsan Kilic, Aurélien Desoeuvres, Frederik Christensen, Deniz Kenan Kilic, and Peter Nielsen. Novel q-learning-based neighborhood discovery algorithm for directed antennas. *Array*, 30:100827, 2026. <https://doi.org/10.1016/j.array.2026.100827>.
- [3] Andreas Kühne Larsen, Kemal Ihsan Kilic, Magnus Berg Ladefoged, and Inkyung Sung. A Novel Path-finding Approach for Maritime Search-and-Rescue Missions Incorporating Dynamic Probability of a Target Location. *Engineering Optimization*, 0(0):1–20, 2025. <https://doi.org/10.1080/0305215X.2024.2446590>.
- [4] Kemal Ihsan Kilic, Samir Maity, Inkyung Sung, and Peter Nielsen. Challenges and AI-Driven Solutions in Maritime Search and Rescue Planning: A Comprehensive Literature Review. *Marine Policy*, 178, 2025. <https://doi.org/10.1016/j.marpol.2025.106692>.
- [5] Kemal Ihsan Kilic and Leonardo Mostarda. Novel Concave Hull-Based Heuristic Algorithm For TSP. *Operations Research Forum*, 3(2):25, 2022. <https://doi.org/10.1007/s43069-022-00137-9>.

- [6] Kemal Ihsan Kilic and Leonardo Mostarda. Heuristic Drone Pathfinding Over Optimized Charging Station Grid. *IEEE Access*, 9:164070–164089, 2021. <https://doi.org/10.1109/ACCESS.2021.3134459>.
- [7] Kemal Ihsan Kilic and Leonardo Mostarda. Optimum Path Finding Framework for Drone Assisted Boat Rescue Missions. In Leonard Barolli, Isaac Woungang, and Tomoya Enokido, editors, *Advanced Information Networking and Applications*, pages 219–231. Springer International Publishing, Cham, 2021. [https://doi.org/10.1007/978-3-030-75078-7\\_23](https://doi.org/10.1007/978-3-030-75078-7_23).
- [8] Kemal Ihsan Kilic, Orhan Gemikonakli, and Leonardo Mostarda. Voronoi Tessellation-based load-balanced multi-objective priority-based heuristic optimisation for multi-cell region coverage with UAVs. *International Journal of Web and Grid Services*, 17(2):152–178, 2021. <https://doi.org/10.1504/IJWGS.2021.114574>.
- [9] K. I. Kilic, O. Gemikonakli, and L. Mostarda. Multi-objective Priority Based Heuristic Optimization for Region Coverage with UAVs. In L. Barolli, F. Amato, F. Moscato, T. Enokido, and M. Takizawa, editors, *Advanced Information Networking and Applications*, pages 768–779. Springer International Publishing, 2020. [https://doi.org/10.1007/978-3-030-44041-1\\_68](https://doi.org/10.1007/978-3-030-44041-1_68).
- [10] F. Al-Turjman and K.I. Kilic. Energy-Aware Routing Protocol for Nanosensor Networks. In Fadi Al-Turjman, editor, *Internet of Nano-Things and Wireless Body Area Networks (WBAN)*, chapter 9. CRC Press, Boca Raton, FL, 2019. <https://doi.org/10.1201/9780429243707>.
- [11] F. Al-Turjman and K.I. Kilic. Lagoon: A simple energy-aware routing protocol for wireless nano-sensor networks. *IET Wireless Sensor Systems*, 9(3):110–118, 2019. <https://doi.org/10.1049/iet-wss.2018.5079>.
- [12] Kemal Ihsan Kilic and Rahib Hidayat Abiyev. Exploiting the synergy between fractal dimension and lacunarity for improved texture recognition. *Signal Processing*, 91(10):2332–2344, 2011. <https://doi.org/10.1016/j.sigpro.2011.04.018>.
- [13] Rahib Hidayat Abiyev and Kemal Ihsan Kilic. Robust Feature Extraction and Iris Recognition for Biometric Personal Identification. In Zahid Riaz, editor, *Biometric Systems*, chapter 9. IntechOpen, Rijeka, 2011. <https://doi.org/10.5772/18374>.
- [14] Rahib Abiyev and Kemal Ihsan Kilic. An efficient fractal measure for image texture recognition. In *2009 Fifth International Conference on Soft Computing, Computing with Words and Perceptions in System Analysis, Decision and Control*, pages 1–4, 2009. <https://doi.org/10.1109/ICSCCW.2009.5379454>.
- [15] Rahib Abiyev and Kemal Kilic. Adaptive Iris Segmentation. In Jong Hyuk Park, Hsiao-Hwa Chen, Mohammed Atiquzzaman, Changhoon Lee, Tai-hoon Kim, and Sang-Soo Yeo, editors, *Advances in Information Security and Assurance*, pages 90–99, Berlin, Heidelberg, 2009. Springer Berlin Heidelberg. [https://doi.org/10.1007/978-3-642-02617-1\\_10](https://doi.org/10.1007/978-3-642-02617-1_10).
- [16] Eenjun Hwang, Kemal Kilic, and V. S. Subrahmanian. Handling Updates and Crashes in VoD Systems. *Multimedia Tools and Applications*, 7:103–132, 1998. <https://doi.org/10.1023/A:1009626304548>.

